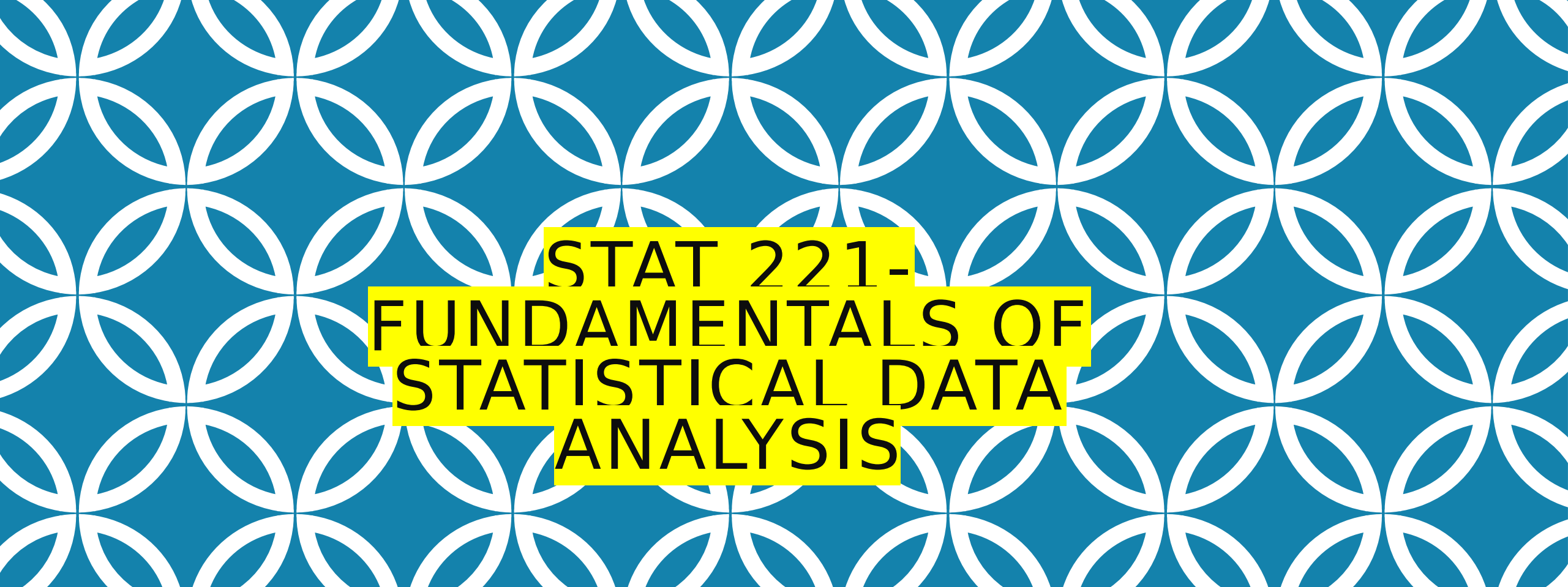
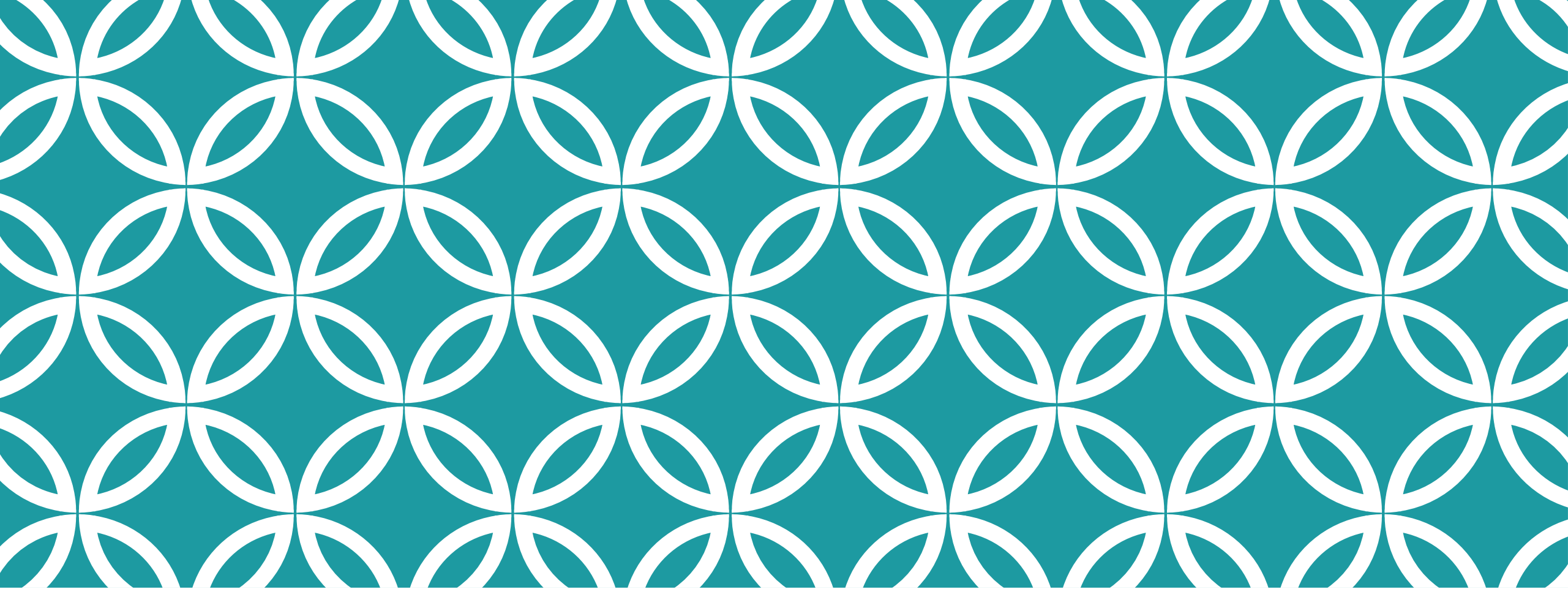




STAT 221- FUNDAMENTALS OF STATISTICAL DATA ANALYSIS

Week-5: Measures of Central Tendency,
Dispersion and Distributions

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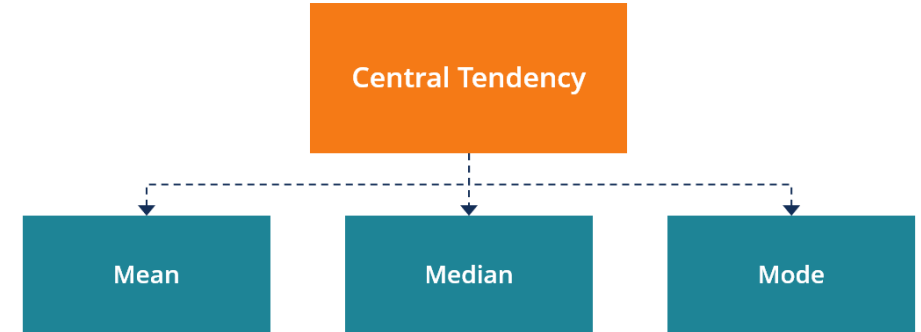


MEASURES OF CENTRAL TENDENCY |

MEASURES OF CENTRAL TENDENCY

TABLE 1 *Symbols and Formulas*

TERM/SYMBOL	MEANING OF SYMBOL	FORMULA
Measures of central tendency	A group of statistics (mode, median, and mean) that are used to describe where cases tend to fall in a distribution. They tell us what is typical in a distribution.	
Mode (Mo)	The most frequently occurring value or attribute	
Median (Md)	The value of the middle case in a rank-ordered distribution	
Mean ($\bar{X}$)	The mathematical average for sample data	$\bar{X} = \frac{\sum X}{N}$
Mean (μ)	The mathematical average for population data	$\mu = \frac{\sum X}{N}$
Position of the median	Formula used to determine which case number (in a rank-ordered distribution of cases) is the median	$\frac{N + 1}{2}$
Outliers	Cases with very high or very low values	



Median The value of the middle case in a rank-ordered distribution.

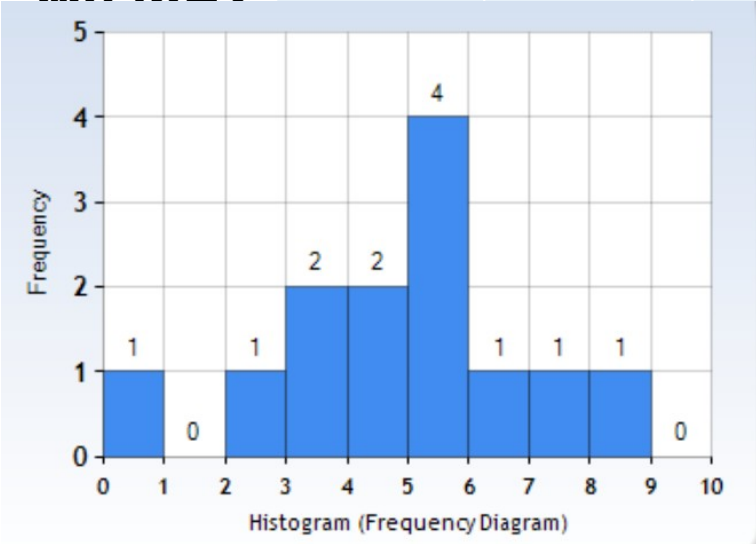
$$\text{Position of the median} = \frac{N + 1}{2}$$

$$\text{Mean} = \frac{\sum X_i}{N}$$

MEAN, MEDIAN, MODE OF INDIVIDUAL DATA

RIDE R #	# BUS TRIPS
1	5
2	3
3	10
4	5
5	8
6	6
7	5
8	4
9	0
10	2
11	14
12	5
13	7
14	4
15	3
N=15	81

Mode=?
Median=?
Mean=?



Median $(N+1)/2$
Median no of rides

No of rides	FREQUENCY (no of people)	PERCENT	CUMULATIVE PERCENT
0	1	6.7	6.7
2	1	6.7	13.3
3	2	13.3	26.7
4	2	13.3	40.0
5	4	26.7	66.7
6	1	6.7	73.3
7	1	6.7	80.0
8	1	6.7	86.7
10	1	6.7	93.3
14	1	6.7	100.0
Total	15	100.0	

WHEN TO USE MEAN, MEDIAN, AND MODE

Religious affiliation	Frequency	Percentage
Hindu	86	57.33
Muslim	22	14.67
Sikhs	12	8.00
Jain	11	7.33
Christian	6	4.00
Atheist	10	6.67
Others	3	2.00
Total	150	100.00

Mode=?
Median=?
Mean=?

How much do you like mess food?	Frequency
Not all	15
Somewhat	23
Very much	12
Total	50

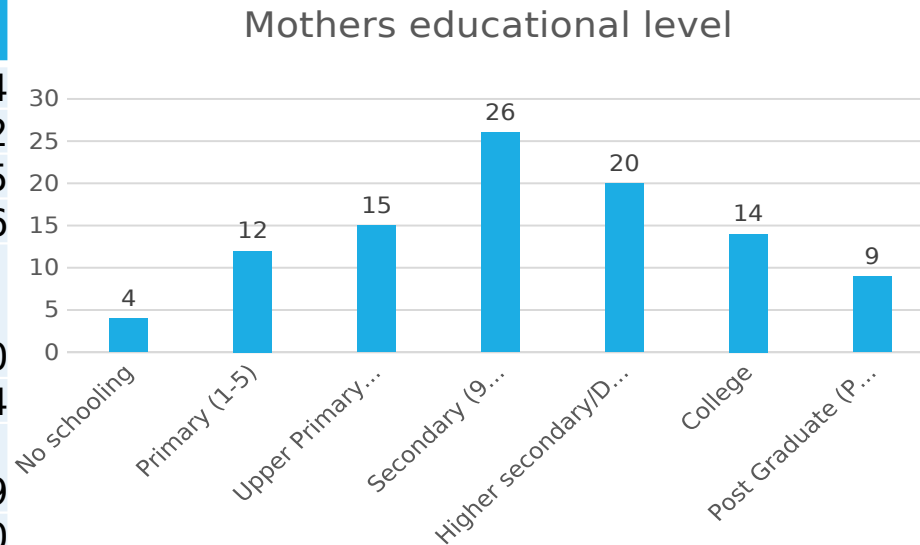
Mode=?
Median=?
Mean=?

TABLE 2 *When to Use the Mode, Median, and Mean*

	NOMINAL	ORDINAL	INTERVAL/RATIO
Mode	✓	✓	✓
Median	NA	✓	✓
Mean	NA	NA	✓

ORDINAL VARIABLES

Mothers education level (attended)	Frequency
No schooling	4
Primary (1-5)	12
Upper Primary(6-8)	15
Secondary (9-10)	26
Higher secondary/Diploma/Vocational	20
College	14
Post Graduate (PG and above)	9
Total	100

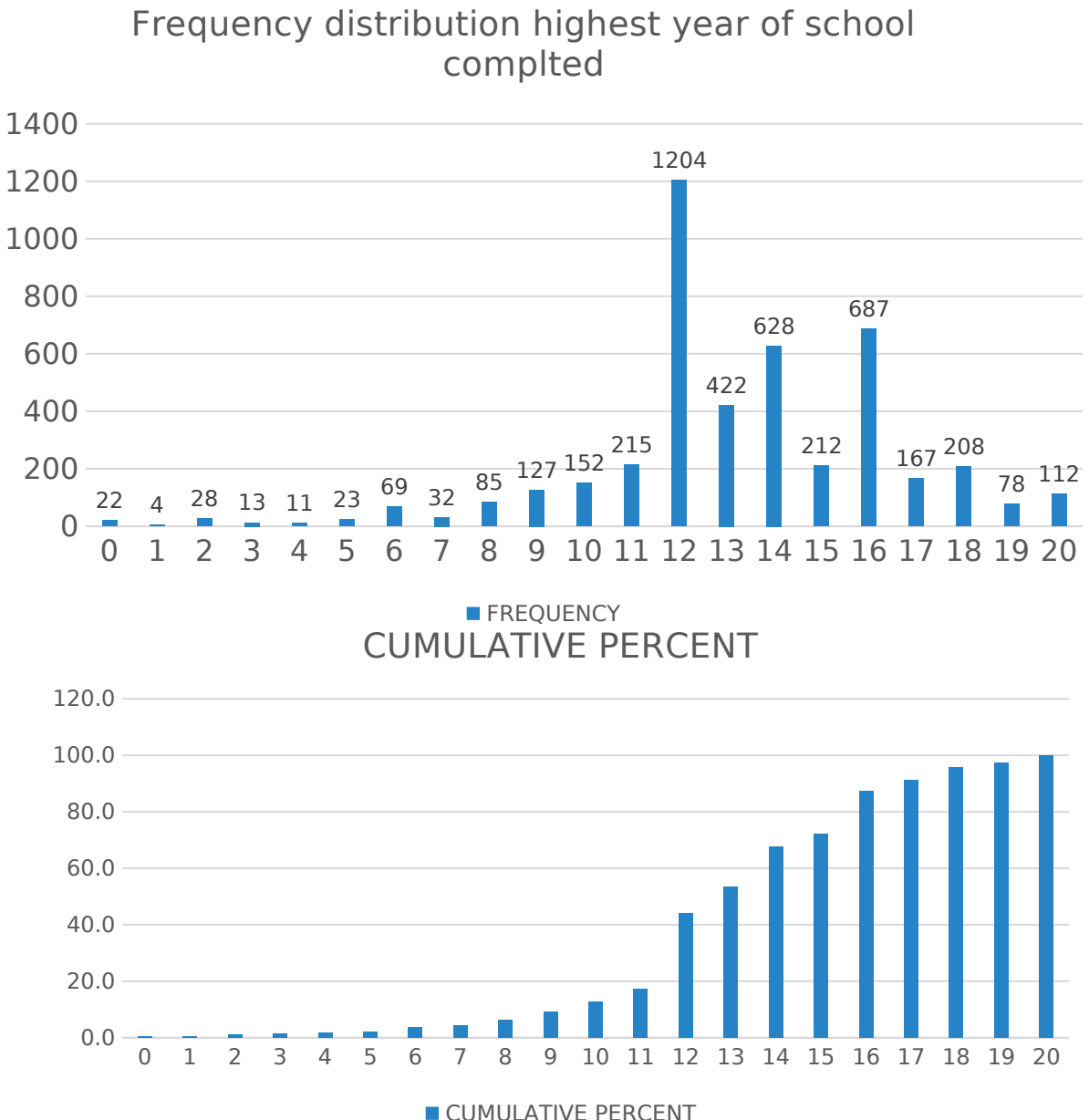


Mothers education level (attended)	Frequency	Cumulative Frequency
No schooling	4	4
Primary (1-5)	12	16
Upper Primary(6-8)	15	31
Secondary (9-10)	26	57
Higher secondary/Diplo ma/Vocational	20	77
College	14	91
Post Graduate (PG and above)	9	100
Total	100	

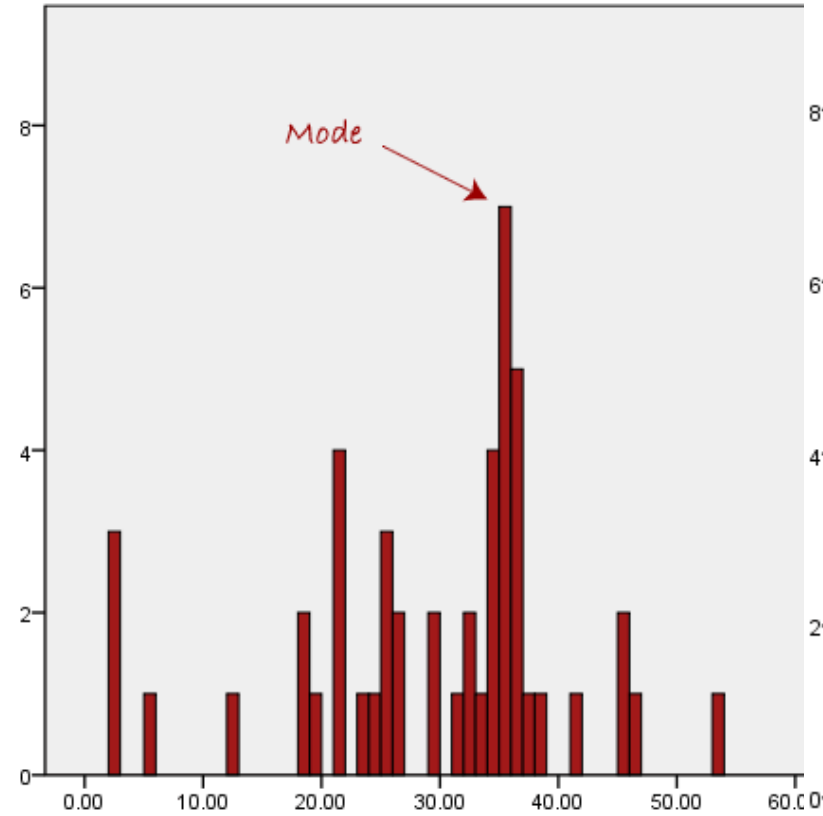
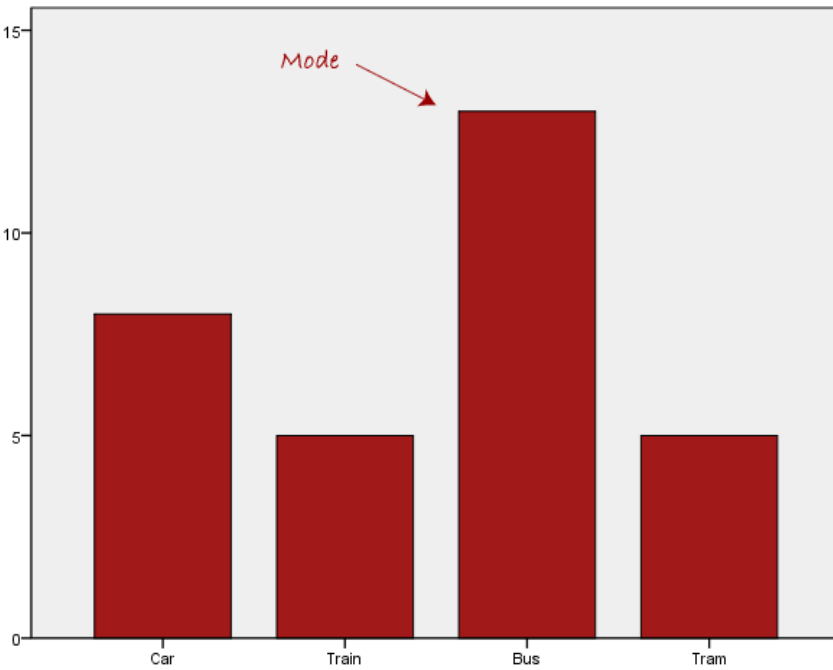
Mode=
Median=
Mean=?

Highest year of school completed	FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
0	22	0.5	0.5	0.5
1	4	0.1	0.1	0.6
2	28	0.6	0.6	1.2
3	13	0.3	0.3	1.5
4	11	0.2	0.2	1.7
5	23	0.5	0.5	2.2
6	69	1.5	1.5	3.8
7	32	0.7	0.7	4.5
8	85	1.9	1.9	6.4
9	127	2.8	2.8	9.2
10	152	3.4	3.4	12.6
11	215	4.8	4.8	17.4
12	1204	26.7	26.8	44.1
13	422	9.4	9.4	53.5
14	628	13.9	14.0	67.5
15	212	4.7	4.7	72.2
16	687	15.2	15.3	87.4
17	167	3.7	3.7	91.2
18	208	4.6	4.6	95.8
19	78	1.7	1.7	97.5
20	112	2.5	2.5	100.0
Total	4499	99.8	100.0	

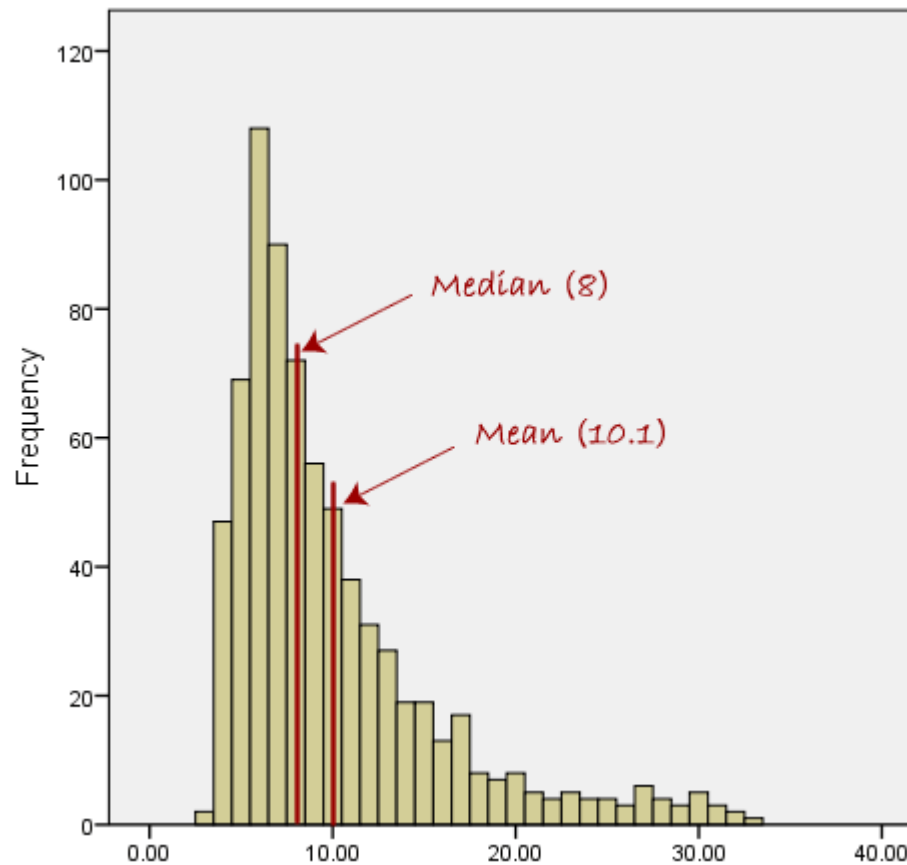
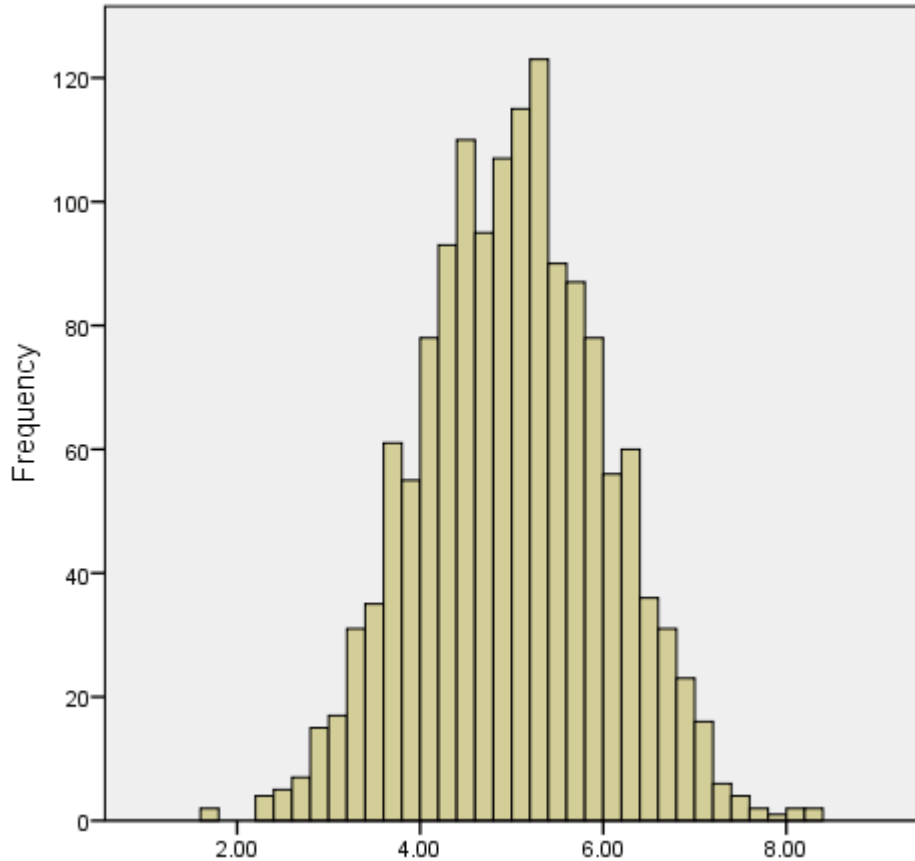
INTERVAL/RATIO- GROUPED DATA



MODE



SKEWED DISTRIBUTIONS AND THE MEAN AND MEDIAN

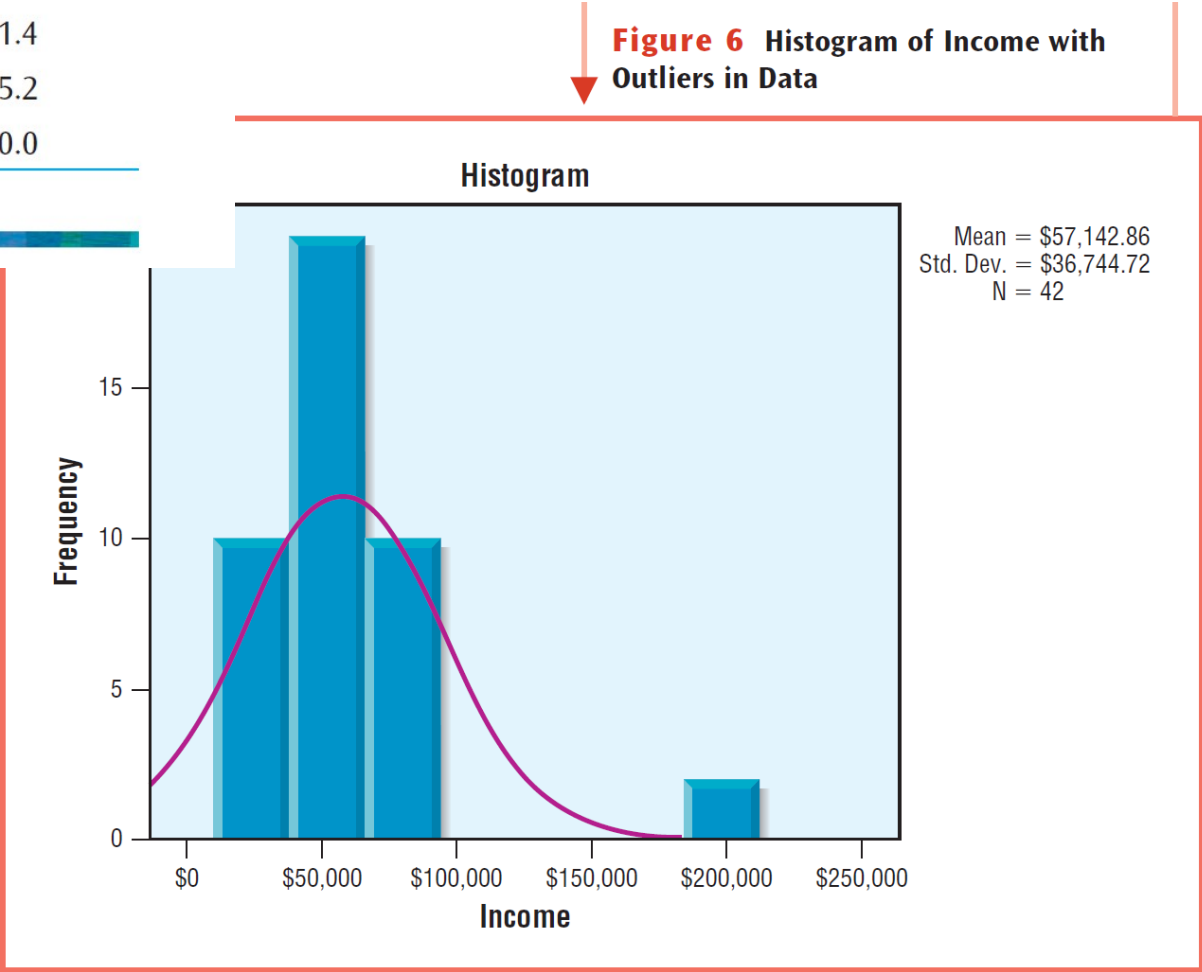


<https://statistics.laerd.com/statistical-guides/measures-central-tendency-mean-mode-median.php>

TABLE 12 *Income*

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	\$25,000	10	23.8	23.8	23.8
	\$50,000	20	47.6	47.6	71.4
	\$75,000	10	23.8	23.8	95.2
	\$200,000	2	4.8	4.8	100.0
Total		42	100.0	100.0	

Figure 6 Histogram of Income with Outliers in Data



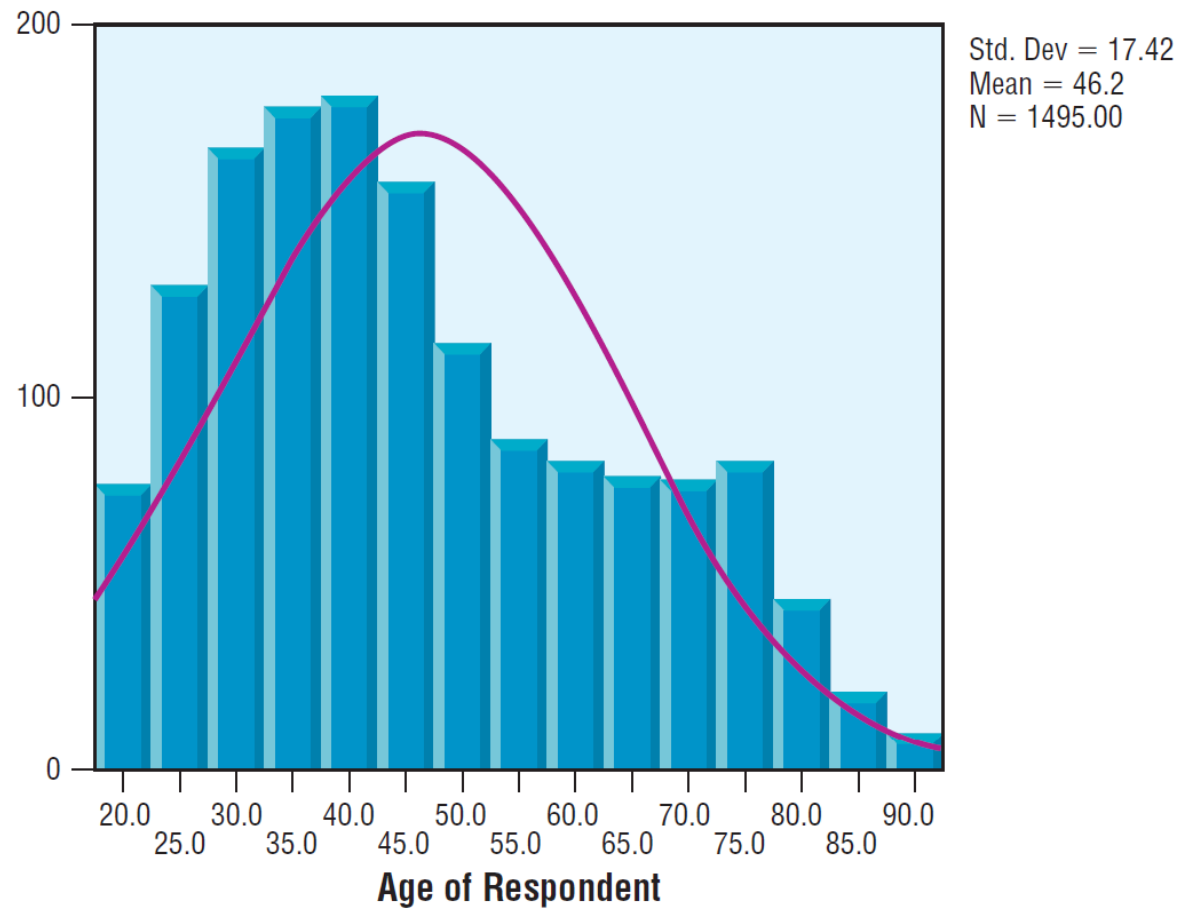


Figure 7 Histogram of Age of Respondent

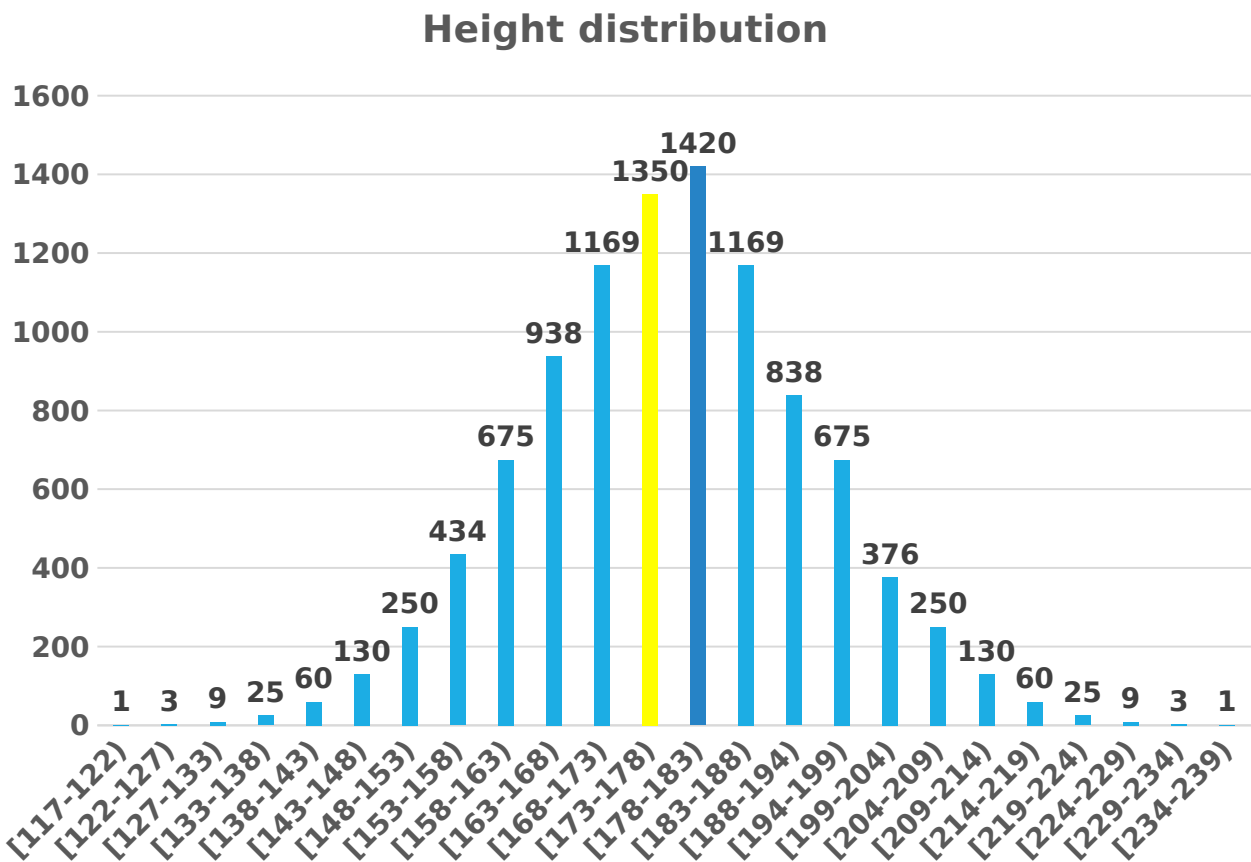
EXERCISE

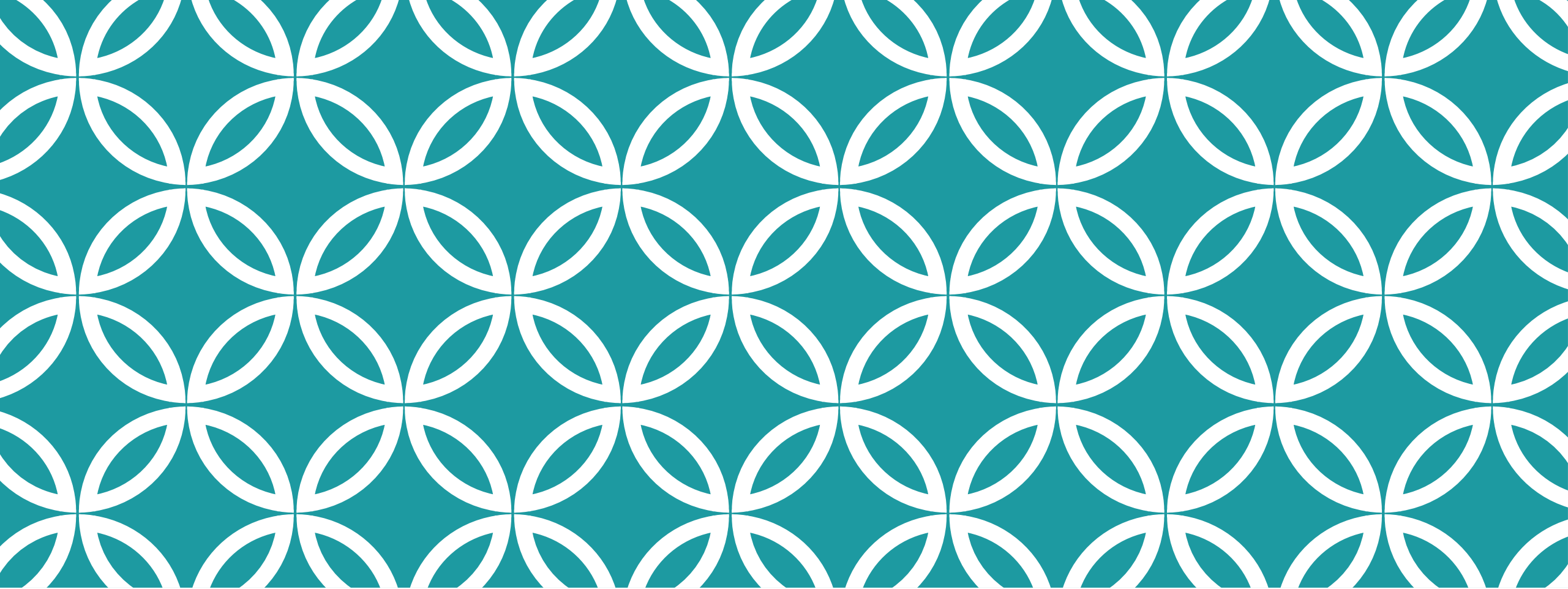
No of books read last one year	Frequency	Cumulative Frequency	Percentage
None	7		
1-2 books	10		
3-5 books	11		
6-10 books	8		
11 or more books	8		
Total	44		
Mode	?		
Median	?		

No of brothers/sisters	Frequency	Percentage	Cumulative percentage	Siblings-x-frequency
0	4			
1	15			
2	11			
3	8			
4	6			
5	6			
Total	50			
	Median			
	Mode			
	Mean			

Height Interval (cm)	Frequency	Cumulative frequency	Percentile		Height- middle	Height times frequency
[117-122)	1					
[122-127)	3					
[127-133)	9					
[133-138)	25					
[138-143)	60					
[143-148)	130					
[148-153)	250					
[153-158)	434					
[158-163)	675					
[163-168)	938					
[168-173)	1169					
[173-178)	1350					
[178-183)	1420					
[183-188)	1169					
[188-194)	838					
[194-199)	675					
[199-204)	376					
[204-209)	250					
[209-214)	130					
[214-219)	60					
[219-224)	25					
[224-229)	9					
[229-234)	3					
[234-239)	1					
Total	10000					
					SUM	?
Median Height	?				MEAN	?
Mode	?					

Height Interval	Frequency	Cumulative	Percentile		Height-mid	Height time
[117-122)	1	1	0.01		119.5	119.5
[122-127)	3	4	0.04		124.5	373.5
[127-133)	9	13	0.13		130	1170
[133-138)	25	38	0.38		135.5	3387.5
[138-143)	60	98	0.98		140.5	8430
[143-148)	130	228	2.28		145.5	18915
[148-153)	250	478	4.78		150.5	37625
[153-158)	434	912	9.12		155.5	67487
[158-163)	675	1587	15.87		160.5	108337.5
[163-168)	938	2525	25.25		165.5	155239
[168-173)	1169	3694	36.94		170.5	199314.5
[173-178)	1350	5044	50.44		175.5	236925
[178-183)	1420	6464	64.64		180.5	256310
[183-188)	1169	7633	76.33		185.5	216849.5
[188-194)	838	8471	84.71		191	160058
[194-199)	675	9146	91.46		196.5	132637.5
[199-204)	376	9522	95.22		201.5	75764
[204-209)	250	9772	97.72		206.5	51625
[209-214)	130	9902	99.02		211.5	27495
[214-219)	60	9962	99.62		216.5	12990
[219-224)	25	9987	99.87		221.5	5537.5
[224-229)	9	9996	99.96		226.5	2038.5
[229-234)	3	9999	99.99		231.5	694.5
[234-239)	1	10000	100.00		236.5	236.5
Total	10000					
					SUM	1779560
Median Height	175.5				MEAN	178.0
MODE	180.5					





FACTFULNESS
CONTROL DRAMATIC
INSTINCTS IN NUMBERS



CLASS QUESTIONS

1. In all low-income countries across the world today, how many girls finish primary school?

- ☐ A: 20 percent
- ☐ B: 40 percent
- ☐ C: 60 percent

2. Where does the majority of the world population live?

- ☐ A: Low-income countries
- ☐ B: Middle-income countries
- ☐ C: High-income countries

3. In the last 20 years, the proportion of the world population living in extreme poverty has ...

- ☐ A: almost doubled
- ☐ B: remained more or less the same
- ☐ C: almost halved

4. What is the life expectancy of the world today?

- ☐ A: 50 years
- ☐ B: 60 years
- ☐ C: 70 years

6. The UN predicts that by 2100 the world population will have increased by another 4 billion people. What is the main reason?

- ☐ A: There will be more children (age below 15)
- ☐ B: There will be more adults (age 15 to 74)
- ☐ C: There will be more very old people (age 75 and older)

7. How did the number of deaths per year from natural disasters change over the last hundred years?

- ☐ A: More than doubled
- ☐ B: Remained about the same
- ☐ C: Decreased to less than half

9. How many of the world's 1-year-old children today have been vaccinated against some disease?

- ☐ A: 20 percent
- ☐ B: 50 percent
- ☐ C: 80 percent

10. Worldwide, 30-year-old men have spent 10 years in school, on average. How many years have women of the same age spent in school?

- ☐ A: 9 years
- ☐ B: 6 years
- ☐ C: 3 years

11. In 1996, tigers, giant pandas, and black rhinos were all listed as endangered. How many of these three species are more critically endangered today?

- ☐ A: Two of them
- ☐ B: One of them
- ☐ C: None of them

12. How many people in the world have some access to electricity?

- ☐ A: 20 percent
- ☐ B: 50 percent
- ☐ C: 80 percent

13. Global climate experts believe that, over the next 100 years, the average temperature will ...

- ☐ A: get warmer
- ☐ B: remain the same
- ☐ C: get colder